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Math Maestro

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JEE Mains 2020 Jan Chapter wise Question Bank

Vector Algebra

Q1

Let $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

$\vec{a}$ lies in plane of $\vec{b}$ & $\vec{c}$

$$\vec{b} = \hat{i} + \hat{j} \text{ \& \; } \vec{c} = \hat{i} - \hat{j} + 4\hat{k}$$

of $\vec{a}$ bisectors angle between $\vec{b}$ & $\vec{c}$, then

माना $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

तथा $\vec{a}$ सदिश $\vec{b}$ और $\vec{c}$ के समतल में है

जहां $\vec{b} = \hat{i} + \hat{j}$ और $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$ है $\vec{b}$ और $\vec{c}$ के मध्य कोण अर्द्धक $\vec{a}$ है तब

(1) $\vec{a} \cdot \hat{k} + 2 = 0$ (2) $\vec{a} \cdot \hat{k} + 4 = 0$ (3) $\vec{a} \cdot \hat{k} - 2 = 0$ (4) $\vec{a} \cdot \hat{k} + 5 = 0$

7th Jan Morning

Sol

(1)

angle bisector can be कोण अर्द्धक हो सकता है $\vec{a} = \lambda(\vec{b} + \vec{c})$ or या $\vec{a} = \mu(\vec{b} - \vec{c})$

$$\vec{a} = \lambda \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} + \frac{\hat{i} + \hat{j} + 4\hat{k}}{3\sqrt{2}} \right) = \frac{\lambda}{3\sqrt{2}} [3\hat{i} + 3\hat{j} + \hat{i} - \hat{j} + 4\hat{k}] = \frac{\lambda}{3\sqrt{2}} [4\hat{i} + 2\hat{j} + 4\hat{k}]$$

Compare with से तुलना करने पर $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

02

If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$

then $(\lambda, \vec{d}) =$

यदि $\vec{a}, \vec{b}, \vec{c}$ इकाई सदिश इस प्रकार है कि $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ तथा $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ और $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$

तब $(\lambda, \vec{d}) =$

(1) $\left(\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$ (2) $\left(\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$ (3) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$ (4) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$

7th Jan Evening

Sol

(4)

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 0$$

$$3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = \frac{-3}{2}$$

$$\Rightarrow \lambda = \frac{-3}{2}$$

$$\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times (-\vec{a} - \vec{b}) + (-\vec{a} - \vec{b}) \times \vec{a}$$

$$= \vec{a} \times \vec{b} + \vec{a} \times \vec{b} + \vec{a} \times \vec{b}$$

$$\vec{d} = 3(\vec{a} \times \vec{b})$$

03

If volume of parallelopiped whose coterminous edges are $\vec{u} = \hat{i} + \hat{j} + \lambda \hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} + \hat{k}$ & $\vec{w} = \hat{i} + \hat{j} + 3\hat{k}$ is 1 cubic unit then cosine of angle between $\vec{u}$ and $\vec{v}$ is

यदि समान्तर षटफलक जिसके आसन्न भुजाएँ $\vec{u} = \hat{i} + \hat{j} + \lambda \hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} + \hat{k}$ व $\vec{w} = \hat{i} + \hat{j} + 3\hat{k}$ है, का आयतन 1 घन इकाई है तब $\vec{u}$ और $\vec{v}$ मध्य कोण का कोज्या मान है-

(1) $\frac{7}{3\sqrt{10}}$

(2) $\frac{7}{6\sqrt{3}}$

(3) $\frac{5}{3\sqrt{3}}$

(4) $\frac{5}{7}$

8th Jan Morning

Sol

$$\pm 1 = \begin{vmatrix} 1 & 1 & \lambda \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix} \Rightarrow -\lambda + 3 = \pm 1 \Rightarrow \lambda = 2 \text{ or } \lambda = 4$$

For $\lambda = 4$ के लिए

$$\cos \theta = \frac{2+1+4}{\sqrt{6}\sqrt{18}} = \frac{7}{6\sqrt{3}}$$

04

Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ is nonzero vector and $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$, $\vec{a} \cdot \vec{c} = 0$ find $\vec{b} \cdot \vec{c}$

माना $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ तथा $\vec{c}$ एक अशून्य सदिश है तथा $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$, $\vec{a} \cdot \vec{c} = 0$ तो $\vec{b} \cdot \vec{c}$ का मान होगा:

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $-\frac{1}{2}$

(4) $-\frac{1}{3}$

8th Jan Evening

Sol

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(3)

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{a} \times (\vec{b} \times \vec{a})$$

$$-(\vec{a} \cdot \vec{b}) \vec{c} = (\vec{a} \cdot \vec{a}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a}$$

$$-4\vec{c} = 6(\hat{i} - \hat{j} + \hat{k}) - 4(\hat{i} - 2\hat{j} + \hat{k})$$

$$-4\vec{c} = 2\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{c} = -\frac{1}{2}(\hat{i} + \hat{j} + \hat{k})$$

$$\vec{b} \cdot \vec{c} = -\frac{1}{2}$$

05

$$\text{If } \vec{P} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$$

$$\vec{Q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$$

$$\vec{R} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$$

and $\vec{P}$, $\vec{Q}$, $\vec{R}$ are coplanar vectors and $3(\vec{P} \cdot \vec{Q})^2 - \lambda |\vec{R} \times \vec{Q}|^2 = 0$ then value of λ is

9th Jan Morning

Sol

1

$$\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0 \Rightarrow a+1+a+a=0 \Rightarrow a = -\frac{1}{3}$$

$$\vec{P} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$$

$$\vec{Q} = \frac{1}{3}(-\hat{i} + 2\hat{j} - \hat{k})$$

$$\vec{R} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k})$$

$$\vec{P} \cdot \vec{Q} = \frac{1}{9}(-2 - 2 + 1) = -\frac{1}{3}$$

$$\vec{R} \times \vec{Q} = \frac{1}{9} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix} = \frac{1}{9} (\hat{i}(4-1) - \hat{j}(-2-1) + \hat{k}(1+2))$$

$$= \frac{1}{9} (3\hat{i} + 3\hat{j} + 3\hat{k}) = \frac{\hat{i} + \hat{j} + \hat{k}}{3}$$

Let $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 5$, $\vec{b} \cdot \vec{c} = 10$ angle between $\vec{b}$ & $\vec{c}$ equal to $\frac{\pi}{3}$

If $\vec{a}$ is perpendicular to $\vec{b} \times \vec{c}$ then find the value of $|\vec{a} \times (\vec{b} \times \vec{c})|$

9th Jan Evening

Sol

30

$$\vec{b} \cdot \vec{c} = 10 \Rightarrow |\vec{b}| |\vec{c}| \cos\left(\frac{\pi}{3}\right) = 10 \Rightarrow 5 \cdot |\vec{c}| \cdot \frac{1}{2} = 10 \Rightarrow |\vec{c}| = 4$$

Also तथा, $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$

$$|\vec{a} \times (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| \sin\left(\frac{\pi}{2}\right)$$

$$\sqrt{3} \times |\vec{b}| |\vec{c}| \sin\frac{\pi}{3} \times 1 = \sqrt{3} \times 5 \times 4 \times \frac{\sqrt{3}}{2} = 30$$



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